Total Printed Pages: 02

Roll No. [redacted]

Subject Code: ECA 101/ EIA 101
Examination – Dec. - 2025
B.Tech. I Semester :EC/EI
Subject Name: Basic Electronics

Time : 03 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a, b & c are compulsory while Part d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:
Part a – Max 50 words, Part b – Max 50 words,
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

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Q.1	(a)	What are the charge carriers in semiconductors? explain.	02	2	1
	(b)	How majority charge-carrier produced in Si wafer doped with n-type impurities. Draw its characteristics.	02	2	1
	(c)	What is Zener breakdown? What are its applications?	03	1	1
	(d)	What is Tunneling? Draw and explain its working and V-I characteristics.	05	3	1
		OR			
		What do you mean by rectification? Draw and explain the most practical circuit of the full-wave rectifier.	05	3	4
Q.2	(a)	How many working regions/modes are available in Bipolar Junction Transistor?	02	1	2
	(b)	What is the purpose of doping the base region in a Bipolar junction transistor (BJT) at a lower level compared to the collector region?	02	3	2
	(c)	What do you understand by α and β in relation to transistor?	03	2	4
	(d)	Explain the circuit diagram and working of a emitter- follower.	05	3	2
		OR			
		Draw and discuss the input and output characteristics of a n-p-n transistor.	05	3	2

Q.3	(a)	What are the Z-parameters?	02	1	3
	(b)	Mention the benefits of the h-parameters.	02	1	3
	(c)	The input impedance to a common-emitter transistor amplifier is 1.2 kΩ with β = 220, r_o = 10 kΩ and load resistance is 2.7 kΩ. calculate the value of r_e.	03	3	3
	(d)	Derive the h-parameters of a two port network.	05	3	3
		OR			
		Draw and discuss the r_e model of the common-emitter transistor amplifier.	05	3	3
Q.4	(a)	What is the purpose of a RC coupling in multistage amplifier?	02	1	2
	(b)	Differentiate single stage and multistage stage amplifiers.	02	1	2
	(c)	List the different types of coupling used in multistage amplifiers.	03	1	2
	(d)	Draw and discuss the two-stage RC coupled amplifier. Mention the limitations and applications of the RC coupled amplifier.	05	2	2
		OR			
		Three amplifier stages are connected in cascade with 0.05V peak to peak input providing 150V peak-to-peak output. If the voltage gain of first stage is 20 and input to third stage is 15V peak-to-peak determine the overall voltage gain.	05	3	2
Q.5	(a)	List the components of a tuned amplifier?	02	1	2
	(b)	Draw the block diagram of a double tuned amplifier.	02	2	2
	(c)	What are the advantages and disadvantages of the single tuned amplifiers.	03	1	2
	(d)	Compare the performances of a Class B and Class C power amplifier.	05	3	2
		OR			
		Discuss the Class A power amplifier with the help of a circuit diagram.	05	3	2

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